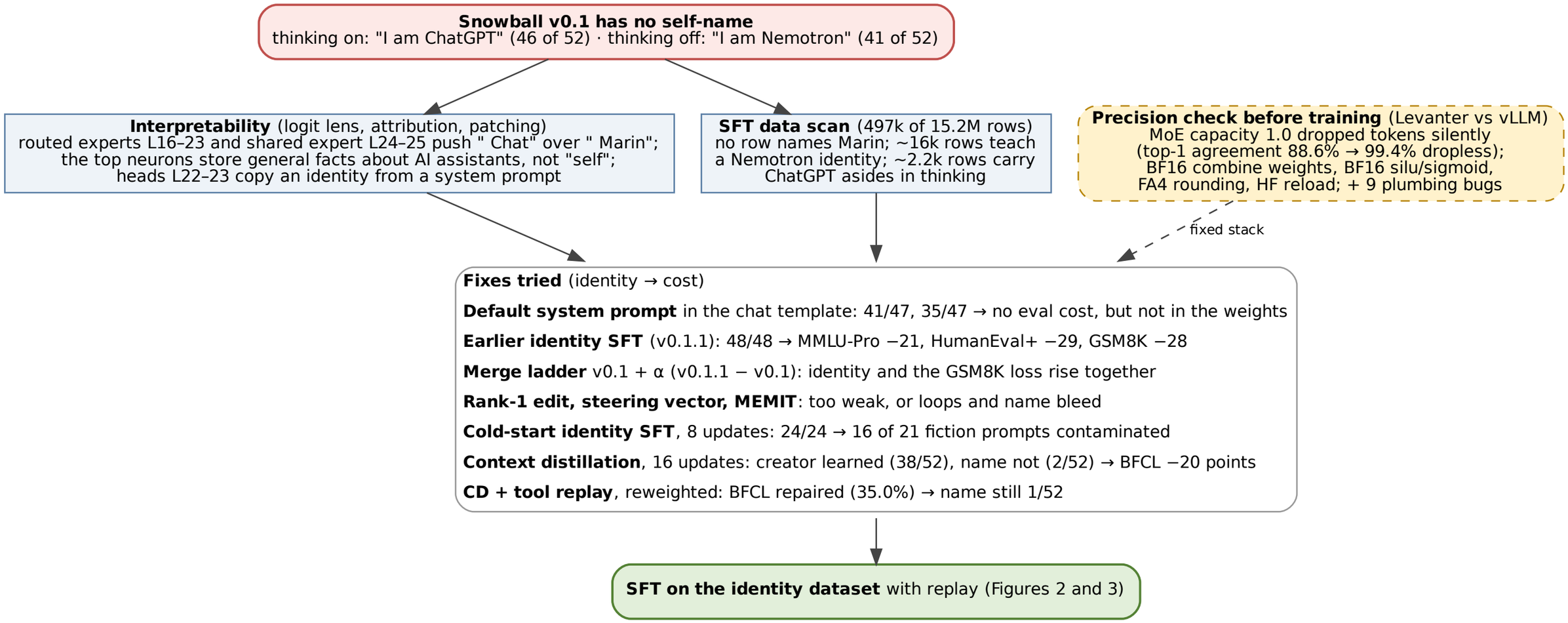
Snowball v0.1 has no self-name
thinking on: "I am ChatGPT" (46 of 52) · thinking off: "I am Nemotron" (41 of 52)
Interpretability (logit lens, attribution, patching)
routed experts L16–23 and shared expert L24–25 push " Chat" over " Marin";
the top neurons store general facts about AI assistants, not "self";
heads L22–23 copy an identity from a system prompt
SFT data scan (497k of 15.2M rows)
no row names Marin; ~16k rows teach
a Nemotron identity; ~2.2k rows carry
ChatGPT asides in thinking
Precision check before training (Levanter vs vLLM)
MoE capacity 1.0 dropped tokens silently
(top-1 agreement 88.6% → 99.4% dropless);
BF16 combine weights, BF16 silu/sigmoid,
FA4 rounding, HF reload; + 9 plumbing bugs
fixed stack
Fixes tried (identity → cost)
Default system prompt in the chat template: 41/47, 35/47 → no eval cost, but not in the weights
Earlier identity SFT (v0.1.1): 48/48 → MMLU-Pro −21, HumanEval+ −29, GSM8K −28
Merge ladder v0.1 + α (v0.1.1 − v0.1): identity and the GSM8K loss rise together
Rank-1 edit, steering vector, MEMIT: too weak, or loops and name bleed
Cold-start identity SFT, 8 updates: 24/24 → 16 of 21 fiction prompts contaminated
Context distillation, 16 updates: creator learned (38/52), name not (2/52) → BFCL −20 points
CD + tool replay, reweighted: BFCL repaired (35.0%) → name still 1/52
SFT on the identity dataset with replay (Figures 2 and 3)